Konvence

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Jednotky

→ používáme jednotky $c = \hbar = 1$

→ veličiny měříme v mocninách eV

$$[A] = (eV)^{\dim[A]}$$

- $\dim[S] = 0$
- $\dim[x] = \dim[t] = -1$
- $\dim[m] = \dim[E] = 1$
- $\dim[J] = 0$
- $\dim[\mathcal{L}] = \dim[\mathcal{H}] = 4$

Lorentzovský indexový formalismus

→ prostoročasové indexy

$\mu, \nu, \alpha, \beta, \dots \in \{0,1,2,3\}$

→ prostorové indexy

$i, j, k, l, \dots \in \{1,2,3\}$

→ Minkowského souřadnice

$$x^\mu \equiv x \equiv (t, \vec{x}) \equiv (x^0, x^i)$$

→ Minkowského metrika

$$\eta_{\mu\nu} = \mathrm{diag}(1,-1,-1,-1) = \eta^{\mu\nu}$$

→ derivace

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} \equiv (\partial_0, \partial_i) \equiv (\partial_t, \vec{\nabla})$$

$$\partial^\mu \equiv \frac{\partial}{\partial x_\mu} \equiv (\partial^0, \partial^i) \equiv (\partial_t, -\vec{\nabla})$$

→ d'Alambertův operátor

$$\Box \equiv \partial_\mu \partial^\mu \equiv \partial_0^2 - \partial_i \partial_i \equiv \partial_t^2 - \Delta$$

→ Levi-Civitův symbol

$$\varepsilon^{\mu\nu\kappa\lambda} = \mathrm{sign}(\sigma)\, \varepsilon^{\sigma(\mu)\sigma(\nu)\sigma(\kappa)\sigma(\lambda)}$$

$$= -\varepsilon_{\mu\nu\kappa\lambda}$$

$$\varepsilon^{0123} = 1 = -\varepsilon_{0123}$$